Procedure, Not Property: SAE Features Whose Ablation Improves Multilingual Reasoning Are Not (Just) Language Features

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1. The Question

Zhao et al. (NeurIPS 2025)

Multilingual LLMs reason worse in non-English languages. Zhao et al. showed this is partly caused by *language-specific representations interfering with reasoning*. Their method: compute per-language mean hidden states across translated math problems, extract a "language subspace" via SVD, and project it out at inference time ($\hat{h} = h - \lambda M_s M_s^\top h$). Across 10 models and 10 languages: consistent gains up to +12.8pp, especially for low-resource languages.

Our extension

Zhao's method operates on entire subspaces and cannot identify individual features or explain the mechanism. We decompose the effect to individual SAE features to characterize both the **mechanism** and the **nature** of the ablated representations.

The question

What are these features doing, and what are they encoding?

Using SAEs on Gemma 3 4B IT, we identify candidate features via monolinguality (ν) $\cap$ supervised probe. Ablation improves reasoning for en +13.2pp, es +6.8pp — but three lines of evidence suggest the label "language feature" is a **procedural tag**, not an **intrinsic property**.

2. SAE Language Feature Extraction

We pass 250 MGSM math problems $\times$ 5 languages through Gemma 3 4B IT, extract last-token residual-stream activations, and encode them through Gemma Scope 2 SAEs (16k-width, layers {9, 17, 22, 29}).

Two identification methods

- **Method A — Monolinguality (ν):** $\nu_s^L = \mu_s^L - \gamma_s^L$ (mean activation on target language minus mean across other languages). Top-50 per language.
- **Method B — Supervised probe:** Logistic regression on SAE feature vectors (5-way language classification, accuracy ceiling $\sim$0.88). Top-50 per language by coefficient magnitude.

Candidates = A $\cap$ B intersection. Methods mostly disagree (Jaccard 0.03–0.33) — the intersection selects features that *both* methods agree are language-differential.

Identifying reasoning features

MGSM contains the same 250 math problems translated into each language. A feature is tagged as **reasoning** if it activates (above threshold) in *all 5 languages for the same problem* on $\geq$10% of problems — i.e., it responds to the mathematical content, not the language. This reduces 16,384 SAE features to just 19 at layer 17 (0.12%).

3. Ablation Results

A. Downstream accuracy

Ablate top-k candidate features per language at layer 17 using directional (QR-orthogonalized) projection. k=20 selected on 50-prompt dev split.

MGSM accuracy after language-feature ablation (layer 17, k=20)

Lang	Baseline	Zhao SVD	SAE ablation
en	0.580	0.632	**0.712**
zh	0.624	0.624	0.596
es	0.568	0.600	**0.636**
bn	0.564	0.564	0.480
sw	0.320	0.332	0.296
Mean	0.531	0.550	0.544

Mean accuracy under SAE ablation (0.544) is comparable to Zhao's subspace baseline (0.550), confirming that individual-feature ablation replicates the subspace-level effect. SAE ablation outperforms Zhao for English (+8.0pp) and Spanish (+3.6pp), suggesting feature-level targeting can be more precise than subspace projection for some languages.

B. Which features affect performance?

Not all statistically-identified candidates behave alike. We ablate each of the top-5 A$\cap$B features per language individually and measure (i) perplexity delta on target-language text and (ii) accuracy delta on MGSM reasoning. A feature is tagged **LANGUAGE** if ablation raises perplexity but does not hurt accuracy (language-only); **SHARED** if it raises perplexity *and* drops accuracy (language and reasoning entangled).

Lang	LANG	SHARED	Other
en	**5**	0	0
es	**3**	0	2
zh	3	2	0
sw	2	**3**	0
bn	0	**5**	0

English's top-5 are all cleanly LANGUAGE; Bengali's are all SHARED. This directly predicts the asymmetry in 3A: ablating clean language features helps; ablating shared features hurts.

C. What happens to the reasoning features?

We ablate the same language features at L17 and measure how the 21 reasoning features respond (forward-only, 50 problems $\times$ 5 langs, Wilcoxon signed-rank). Neuronpedia look-ups reveal three functional camps:

Camp	Neuronpedia identity	Effect
Reasoning (4094, 5993, 1105)	solving for x, step-by-step, causal reasoning	**Stable** $\lvert\Delta\rvert<200$
Answering (96, 406, 377, 712, 9240)	explanatory, QA patterns, response initiation	**Boosted** Δ up to +7036
Answering (107, 279, ...)	conversational acknowledgment, preamble ("Okay,...")	**Suppressed** to $\approx$0

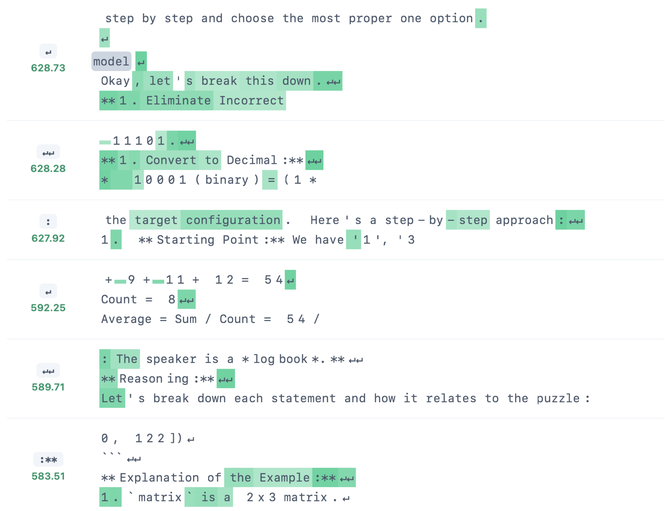

Figure 1. Feature 1105 (L17) on Neuronpedia: fires on causal/sequential reasoning ("let's break this down," "step-by-step"). This genuine reasoning feature is **unaffected** by language-feature ablation (Δ<200).

Ablation reorganizes answering, not reasoning. The genuine reasoning features (4094, 5993, 1105) are **already active and independent of language**. What changes are the *answering* features: explanatory features get boosted while conversational preamble gets suppressed — the model shifts toward more direct answer generation. This explains the en/es asymmetry: clean LANGUAGE features ablate without side effects, reshuffling answer generation; SHARED features (bn/sw) disrupt more than just language, so the improvement doesn't come through.

4. But What Does It Mean to Be a Language Feature?

A. Humanly-interpretable features — through Neuronpedia

We look up every chosen feature on Neuronpedia. Even for **English and Spanish** — our best-performing languages — only 2–3 of the top features are humanly interpretable as language-specific. For **Bengali and Swahili**, *none* map to recognizable language markers.

We hand-selected only Neuronpedia-confirmed features and ablated them (5–6 per language, n=50). **Result: no detectable change in reasoning performance** ($\pm$0–4pp, indistinguishable from random ablation). The causally effective features are the statistically-identified ones that lack clear interpretability.

B. Are these language features global?

Both Zhao's subspace and our SAE features are identified on math content. We run the *same* identification procedure on FLORES-200 (1012 parallel translations, no math) and compare.

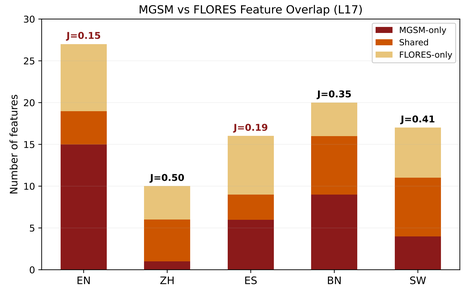

Figure 2. Same procedure, different domain $\rightarrow$ different features. English (J=0.15) and Spanish (J=0.19) share almost none.

The features identified as "language features" on MGSM are largely **different** from those identified on FLORES. A language probe trained on MGSM features transfers poorly to FLORES (accuracy 0.714, need >0.80), and the LANGUAGE/SHARED distinction from Section 3B is meaningless out of domain (p=0.51). If these were genuinely encoding *language identity*, they would generalize across domains. They don't — they encode **task-specific structure that co-varies with language** on a particular dataset.

5. Conclusion

1. **"Language feature" is a method-dependent category.** Every procedure for identifying language features depends on what definition you start from, and no definition is privileged.
2. **The procedure works.** Statistical identification + ablation reproduces and extends Zhao's gains (en +13.2pp, es +6.8pp).
3. **But these are not reliably language features.** They don't generalize out of domain (probe transfer 0.714, separation p=0.51), don't match Neuronpedia auto-interp, and change when the domain changes (en Jaccard=0.15).
4. **"Reasoning feature" is also a procedural tag.** The most dramatic released feature (96, Δ up to +7036) fires on word-problem content tokens, not mathematical operations. Both labels are method-dependent.

Limitations & future work

Single model (Gemma 3 4B IT), single task (MGSM), 5 languages; global k=20 suboptimal; dev/test overlap in profiling step. Next step: compare across feature identification operationalizations (causal LM ablation, top-activating examples, activation-steering).

References

Zhao et al. (NeurIPS 2025); Deng et al. (2025); Bricken et al. (2023); Lieberum et al. (2024)